Chapter 9

Drug Information Services and Sources of Information

Matthew P. Van Cuyk[1], Jason C. Cooper[2] and James P. New[2]
[1]Massachusetts General Hospital, Boston, MA, United States; [2]Medical University of South Carolina, Charleston, SC, United States

Learning Objectives

Objective 9.1 Outline features of drug information centers and types of drug information services provided in a hospital.
Objective 9.2 Describe applications of common sources of drug information.
Objective 9.3 Overview of drug information quality on the Internet.

OBJECTIVE 9.1. OUTLINE FEATURES OF DRUG INFORMATION CENTERS AND TYPES OF DRUG INFORMATION SERVICES PROVIDED IN A HOSPITAL

Drug Information Centers

In August 1962, the first drug information center (DIC) was established in the United States at the University of Kentucky.[1] The primary objective of this DIC was to provide comprehensive drug information services and to play an active role in the education of medical, dentistry, nursing, and pharmacy students.

Since 1962, many DICs have been established around the world (Fig. 9.1). For a more complete list of DICs around the world, the Society of Hospital Pharmacists of Australia (SHPA) publishes a document that contains the contact information for these centers. The document the SHPA publishes includes DICs in 51 countries; however, it is not all-inclusive.[2]

By 1974, there were 56 DICs in the United States, which increased to 108 by 1992.[8,9] However, by 2008 the number of centers in the United States dropped to 77.[10] Today, many DICs in the United States are affiliated with large academic teaching institutions, a college of pharmacy, or both. This decrease may be attributed to increased training in pharmacy curricula regarding the necessary drug information skills for pharmacists. Additionally, there has also been an increase in accessibility to valid medical and pharmaceutical Internet databases to help answer various types of drug information questions. Although DICs in many countries began independently at universities or hospitals, some countries, such as the United Kingdom, have a national organization that helps maintain centers throughout the country.[11]

To start a DIC, an individual trained in the provision of drug information and a computer with Internet access will be needed, at a bare minimum. However, ideally, a DIC would have several trained individuals to help maintain the day-to-day functions of the center and access to at least one resource pertaining to each of the different types of reference categories listed later on in this chapter. In regard to the amount of space needed for a DIC, there is no standard amount; however, enough space is required to house computers, references, and supplies, and for the staff and learners to be comfortable in a quiet work environment. Finally, financing a DIC requires enough money for pharmacists to staff the center and provide access to various references. In determining a DIC budget, it is important to keep in mind that text references may cost hundreds of dollars, whereas access to online databases for an institution may cost thousands of dollars.

In addition to more traditional DICs, drug manufacturers have medication information departments that serve as a source of drug information for the products that the manufacturer produces. Another example of a nontraditional DIC would be poison control or toxicology centers. In the United States, there are 55 Poison Control Centers at the time of preparation of this chapter, which can be reached via telephone at 1-800-222-1222 in any state.[12]

Clinical Pharmacy Education, Practice and Research. https://doi.org/10.1016/B978-0-12-814276-9.00009-X
Copyright © 2019 Elsevier Inc. All rights reserved.

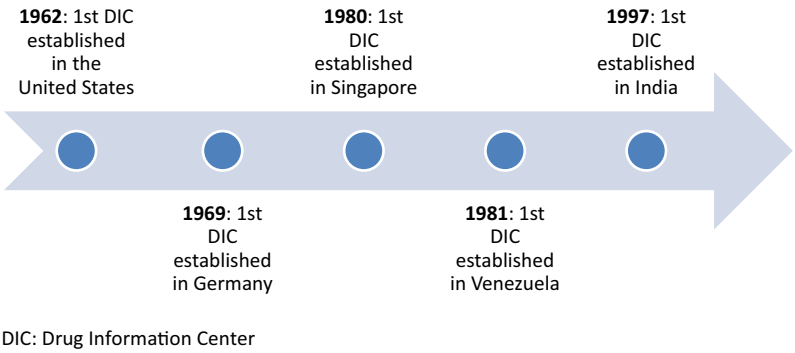


FIGURE 9.1 Examples of first drug information centers established around the world.[1,3–7] *DIC*, drug information center.

DICs participate, in general, in

- Responding to questions
- Formulary management
- Medication use evaluations (MUEs)
- Adverse drug reaction (ADR) reporting
- Communicating newsletters, precepting pharmacy students and pharmacy residents, teaching courses to healthcare professionals in training, and providing in-services to staff.

Responding to Drug Information Questions

One of the primary services provided by DICs, whether it is an academic center, a center associated with a manufacturer, or a poison control center, is to respond to drug/medical questions from healthcare professionals and/or the general public.[10,13] Most academic DICs mainly answer questions from clinicians within their institution; however, others may answer questions from professionals all around their country or state. Others may have policies and procedures in place to respond to questions from patients or the general public. For example, at the Medical University of South Carolina, the Ask-A-Pharmacist web-based service is used to respond to the general public about questions related to their medications (e.g., ADRs, drug−drug interactions). Many retail pharmacies also have their own online services for patients who get their prescriptions filled at that pharmacy to submit a question regarding their medications.

Formulary Management

The cost of healthcare continues to rise. The purchase, use, and monitoring of medications contribute significantly to this increased cost.[15] Although prudent care requires making medical decisions with a patient on an individual basis, streamlining the use of medications for groups of patients has for many years been a method to ensure highly efficacious, safe, and cost-effective utilization.[16–19] As such, many healthcare institutions have a *formulary system* to direct practical use of medications for patients under their care.

A *formulary*, at its essence, is a continually updated list of medications and associated materials approved for use within a particular healthcare institution/country/region. In addition to a list of medications, the formulary consists of policies, guidelines, and other tools designed to support the proper use of these medications. This list should represent the collective clinical judgment of the professionals practicing in that institution. Policies should also be in place to dictate how the use of medications or practices not on the formulary (i.e., nonformulary) should be handled.

The goal of a formulary system is to ensure that the most effective, safe, and cost-effective medications are available for use by practitioners in a particular healthcare facility. At the heart of this review process is the *Pharmacy (or Drug) and Therapeutics (P&T) Committee*. Drug information pharmacists can provide valuable insight to the Committee during the review of medications due to their understanding of evidence-based principles and abilities to find and synthesize information about a drug for Committee members.

Formulary Reviews

A focus of most requests to a P&T Committee involves the review of new medications. Reviews of these agents should be thorough and unbiased including information in the published literature as well as institutional-specific considerations

TABLE 9.1 Formulary Monograph Sections

- General information
  - Brand name
  - Generic name
  - Drug class
- Indications
  - Labeled indications
  - Off-label indications
- Pharmacology
  - Mechanism of action
  - Pharmacokinetics
  - Dosing
- Operations
  - Dosage forms
  - Storage and stability
  - Special handling
- Literature review
  - Clinical trial analysis
  - Grading of evidence
  - Presence in guidelines
- Safety review
  - Contraindications
  - Adverse effects
  - Warnings and precautions
  - Pregnancy and lactation
  - Drug interactions
  - Required monitoring
  - Risk analysis
- Costs
  - Formulary alternatives
  - Reimbursement
  - Pharmacoeconomic analysis
- Conclusion
  - Summary of information
  - Place in therapy
  - Recommendation

(e.g., operational preferences, payment rates, patient acuity). Typically, medications under consideration are reviewed on an individual basis. However, medications that belong to a single class of drugs or treat a similar disease state may require a *class review* to be considered collectively.

An evaluation of a medication can be put together in a drug monograph and submitted to the P&T Committee for review. These monographs should have a standardized format (Table 9.1).

Drug information pharmacists, along with students and residents on rotation, are valuable in creating, evaluating, and editing these monographs. Information is initially gathered on general information (e.g., pharmacology, adverse effects, storage) with the help of common online tertiary databases such as Micromedex, Facts and Comparisons, or Lexicomp. Additionally, the package insert is useful, as this lists the current indications and use of the drug approved by the Food and Drug Administration (FDA). A thorough primary literature review is then necessary to evaluate the available clinical trials for the medication. This can be researched through commonly used secondary databases such as PubMed, SCOPUS, or Medline. Investigation of the studies for the patient population included, outcomes used, and the results obtained should be performed to identify the external validity of the study to one's own institution and patients. The clinical evaluation of efficacy and safety can then be combined with cost and reimbursement to develop general guidance or recommendation for whether or not to add the particular product to the institution's formulary.

Strategies for Maintenance

Once a decision has been made by the P&T Committee, follow-up should be performed to ensure that optimal care is taking place. A MUE or *drug use evaluation* is one tool used to help with continually reviewing a formulary. MUEs are quality improvement tools used to appraise and optimize how medications are managed in the *medication use process* for the purpose of improving patient outcomes.[16,20] MUEs are commonly included as part of the goals/objectives of a pharmacy residency program. These reviews are often led by residents, with the help and guidance of clinicians in that specialized area. Data collection is performed primarily by pharmacy residents, with the ultimate goal of submitting their research for publication. However, the results and information generated are of great importance to the institution in improving their overall efficacy and safety of medication use. These can be presented to the P&T Committee for follow-up and reassessment of previous formulary decisions.

For example, an intravenous immunoglobulin (IVIG) MUE revealed several diagnoses and uses of IVIG in the hospital by various services. Through a literature search, a survey of other similar-sized hospitals, and discussion among specialists who often use IVIG, a list of supported and approved uses was developed. A cost assessment was also performed to estimate expenses saved by adhering to the list. This information was then presented to the P&T Committee and approved. The adopted usage list for IVIG was then built into the electronic medical record where clinicians on designated services could only use IVIG in their patients from the approved list of indications and diagnoses.

Other techniques for optimizing the use of medications on a formulary include the use of generic medications, therapeutic interchange protocols, injectable to oral conversion protocols, automatic discontinuation protocols, and policy and guidelines development.[16–18,21] Use of these tools themselves should also be subject to the continual review and updating required for other parts of the formulary.

Adverse Drug Reactions/Medication Errors

Many DICs affiliated with a hospital may offer services to help track and report ADRs and medication errors.[4,10,13] One tool that is often used for tracking ADRs and medication errors utilizes a trigger tool created by the Institute for Healthcare Improvement (IHI).[22,23] These trigger tools use medications that are commonly administered after an ADR has occurred to reverse the adverse effects of medication (e.g., naloxone for opioids, flumazenil for benzodiazepines, and vitamin K for warfarin). Another method relies on the E-codes available in the supplemental section of the International Classification of Disease 9th and 10th editions.[24,25] These E-codes are used to describe external causes of injury, such as ADRs, which are coded by medical coders. In addition to tracking ADRs, drug information specialists are often tasked with helping to evaluate and set up decision support tools that will alert physicians and pharmacists to drug interactions (i.e., drug–drug interactions, drug–food interactions, drug–allergy interactions, drug–disease interactions).

Education

Many DICs participate in writing or editing newsletters (Fig. 9.2).[4,10,13,26–28] The primary purpose of a newsletter is to educate staff or faculty about new medications recently approved, guideline and policy updates, medication safety issues, and formulary news. In addition to educating readers, data suggest that newsletters may help alter physician prescribing habits in the short term.[26,28,29] Therefore, newsletters may be a useful tool when trying to persuade physicians to use a medication that is on an institution's formulary, which is often the most cost-effective medication.

In-Services for Staff

In addition to writing or editing newsletters, DICs often educate staff or faculty members through in-service programs.[4,10,13] DICs are in a unique position for this service, as they can easily identify the needs of the staff or faculty.[30] If, for example, a DIC notices an influx in questions about a particular disease state or medication, they can prepare an in-service program to educate the appropriate audience. In addition to identifying topics for an in-service program through the questions that a DIC answers, MUEs can be used to identify areas of problematic utilization of a particular medication or class of medications.[31] After these issues and an appropriate alternative or recommendation have been identified, an in-service presentation can be prepared to educate the staff or faculty on the appropriate use of the medication or class of medications.

Precepting Students/Residents on Rotation

Of course, a valuable part of DICs is training students and residents to respond to questions in any type of environment.[13,14] Students and residents are taught about the various tertiary references and secondary databases available for research. An effective and efficient manner of searching, for example, using the modified systematic approach to answering drug information questions, is often taught by using real-life questions pertaining to actual patients. In addtion, case-based scenarios developed from the primary literature or previous calls can be utilized to enforce the modified systematic approach.

OBJECTIVE 9.2. DESCRIBE APPLICATIONS OF COMMON SOURCES OF DRUG INFORMATION

To answer a query using a systematic approach, the order of search follows tertiary to secondary to primary references. This is a general to specific information approach. This section follows the same order.

Tertiary References

Tertiary references/databases are those that contain a collection of information.[32–34] The primary definition of tertiary references is that they do not contain original information. Data and information may be pulled from primary literature sources, but its assembly into one source does/should not declare anything of new or original value.



PHARMACY & THERAPEUTICS

DRUG INFORMATION FOR HEALTHCARE PROFESSIONALS

IS ENDING THE PRESCRIPTION OPIOID EPIDEMIC EVEN POSSIBLE?

Opioids have been used for acute pain relief for more than a century. However, the misuse of prescription opioids in the present day has caused an epidemic.[1-3] As elucidated in a recent article published in the *New England Journal of Medicine*, a key element that highlights the epidemic is the association between an increased prescribing of opioids and opioid-related morbidity and mortality.[1] According to the U.S. Surgeon General, deaths caused by opioid overdose have increased 3-fold since 1999.[3] Overprescribing of opioid analgesics has contributed to the increased use of for medical or nonmedical reasons. It has been estimated that the amount of prescription opioids that was prescribed in 2013 is enough to supply 1 bottle of pills to every adult in the United States.[2] Among all opioid prescriptions, ones written in primary care settings were associated with a higher rate of opioid dependence than those written in a pain clinic setting.[2]

The social and economic burden of opioid misuse and abuse are well-described in the literature. The total societal costs of prescription opioid abuse has been estimated to be $55 billion in 2007. Emergency department and inpatient costs were estimated to be $21 billion in 2009.[4,5] It was only in 2001 that the Joint Commission implemented new pain management standards.[6] These new standards required systematic pain assessment and reassessment during the continuum of care using quantitative measures such as a 10-point pain scale and eliminated "as-needed" or "PRN-range" pain medication orders. The ultimate goal is to optimize opioid prescribing in a safer and more effective way.[6]

In order to tackle the prescription opioid epidemic, the U.S. Department of Health and Human Services (HHS) recommended a multifaceted, harm-reduction approach. In March 2015, the HHS Secretary announced an initiative that focused on 3 priority areas (Table 1).[2]

Table 1. HHS Goals for the Opioid Epidemic

1. Improve prescribing practices
2. Expand access of medication-assisted treatment (MAT) programs
3. Expand the use of naloxone (Narcan®)

To date, federal agencies have implemented policies, grants, and programs to provide resources to counteract the epidemic. The Centers for Disease Control and Prevention (CDC) published guidelines for primary care clinicians on prescribing opioids and provided over $30 million in grants to the states to strengthen Prescription Drug Monitoring Programs (PDMPs).[2] The Substance Abuse and Mental Health Administration (SAMHSA) awarded over $10 million to 11 high-burden states to expand their MAT programs with plans to continue to expand to additional states.[2] These resources are intended to improve the capacity for care of patients with opioid-use

JULY 2017

Ending the Prescription Opioid Epidemic

Barriers to Naloxone Prescribing

Codeine Use in Pediatric Patients

Formulary Update: June 2017

FIGURE 9.2 Part of drug information center newsletter.

Tertiary references are commonly thought of as books, online databases (e.g., Micromedex, UpToDate), and compendiums. Although convenient and easy to use, tertiary texts may be outdated shortly after publication. Therefore, it is always a good idea to check the publication data for a specific book to determine how current the information is. However, because many texts also are now available online or via smartphone apps, databases and references can be updated easily with new information.

Review articles could also be considered tertiary references if they are more narrative in nature. For example, a general review article regarding the current treatment of gout that only summarizes the major information available, but does not put forth any new recommendations, would likely be considered tertiary. However, metaanalyses and systematic reviews that develop a strategy for selection of primary literature, summarize the gathered information, and

run statistics to establish new approaches or analysis would be considered more as primary literature due to the generation of original or new material.

It is up to the reader to determine the quality of the information presented in tertiary references. Author bias, misinterpretation, missing data, or other errors may appear in such sources. That is why it is critical to validate and verify the information found in more than one tertiary reference.

When researching drug information questions, it is important first to categorize the question being asked. Once a question is categorized into a specific subject matter (e.g., ADR, drug−drug interaction, etc.), the search for information can begin with tertiary references that are written to cover that particular theme. Categorizing the question also eliminates extraneous tertiary references that would not be helpful to one's search. Table 9.2 is a fairly comprehensive list, broken down by drug information category, of a variety of tertiary references, databases, texts, and compendiums that are available to answer drug information questions. Of note, some of these online databases require a subscription. Please check with your institution to determine availability and access. Additionally, many of these online databases offer mobile applications for smartphones.

Secondary References

Secondary references are usually electronic databases that list primary literature articles via an index or abstracting service.[32,33,35] Indexing services provide basic information of the article such as author, title, publication, and year. Abstracting services, which are more common, provide the citation of the article along with a curtailed description of what the article is about.

Not all indexing and abstracting services are created the same; therefore, it is important to search in various databases to retrieve the most pertinent and up-to-date articles available. For example, PubMed uses medical subject headings (i.e., MeSH terms) to search and find primary literature articles; however, a secondary database such as International Pharmaceutical Abstracts (IPAs) does not use MeSH terms and has its own indexing language. Both databases can be searched using keywords.

Of note, keywords will provide more current articles than using only MeSH terms in a searching strategy, as articles that were most recently published (e.g., last week) may not have had a chance to be indexed and tagged with MeSH terms. This is commonly referred to as the "lag time"—the period between when an article is published in a journal to when it appears in a secondary database. Most major journals (e.g., JAMA, New England Journal of Medicine) do not experience much of a lag time, as their articles are updated and listed in various secondary databases immediately or before publication. However, pharmacy or specialized journals may have more of a lag time because their content is not prioritized in a particular secondary database.

Additionally, most secondary databases use tools such as Boolean operators to search a database. Operators like "AND," "OR," and "NOT" can be combined with general search terms to yield more or fewer articles. For example, creating a searching strategy such as "drug compounding [MeSH]" AND "aspirin [MeSH]" AND "suspensions [MeSH]" would retrieve articles that dealt with all three subjects (i.e., formulations/recipes for compounding an aspirin suspension). Therefore, "AND" is commonly used to limit searches. However, using the operator "OR" will expand a search to include articles that discuss items that might be closely related. For example, if one wanted to find articles about the rate and administration of human immune globulin via the intravenous route, part of the search strategy should include "infusions, intravenous [MeSH]," OR "injections, intravenous [MeSH]" as articles may not differentiate between intravenous injections or infusions. This would offer a broader search at first, and if too many articles were available with the "OR" operator, each MeSH term could be used in its own search and combined with "immunoglobulins, intravenous [MeSH]." Using "NOT" as an operator would eliminate articles that were indexed by a particular subject heading that might be confused with another.

Finally, most secondary databases can use limits when searching and retrieving articles. Common limits such as language, clinical trial design, species (i.e., humans), publication dates (e.g., within last 5 years), and age ranges are available to help narrow down searches to a particular group of patients or within a certain time frame. For more detailed information on how to perform and execute helpful searches within secondary databases, below is a short list of common databases.

Cumulative Index to Nursing and Allied Health Literature

Cumulative Index to Nursing and Allied Health Literature (CINAHL) (https://health.ebsco.com/products/the-cinahl-database/allied-health-nursing) provides indexing of articles from top nursing and allied health literature and includes such

TABLE 9.2 List of Tertiary References by Category

Books/Printed References	Online References
Adverse Drug Reactions	
• Meyler's Side Effects of Drugs: The International Encyclopedia of Adverse Drug Reactions and Interactions • AHFS Drug Information (ASHP) • Drug Information Handbook (Lexicomp) • Drug Facts and Comparisons • Drug-Induced Diseases: Prevention, Detection, and Management	• Facts and Comparisons eAnswers (see Table 9.3) • Lexicomp Online (see Table 9.4) • MICROMEDEX (see Table 9.5) • Clin-Alert Online (http://cla.sagepub.com/) • DailyMed (http://dailymed.nlm.nih.gov/dailymed/index.cfm) • FDA Adverse Event Reporting System (www.fda.gov) • VigiAccess (www.vigiaccess.org/)
Biostatistics	
• Drug Information: A Guide for Pharmacists • Basic & Clinical Biostatistics	• Drug Information: A Guide for Pharmacists (via AccessPharmacy) • Basic and Clinical Biostatistics (via AccessMedicine)
Chemical Structural Information	
• Goodman and Gilman's: The Pharmacological Basis of Therapeutics • Remington: The Science and Practice of Pharmacy • Merck Index	• Goodman and Gilman's: The Pharmacological Basis of Therapeutics (via AccessMedicine) • ChemIDplus (via TOXNET) • PubChem (through PubMed or NLM Drug Information Portal) • USP-NF Online (http://www.uspnf.com/uspnf/login)
Compounding	
• Trissel's Stability of Compounded Formulations • Extemporaneous Formulations for Pediatric, Geriatric and Special Needs Patients • Pediatric Drug Formulations • Handbook of Extemporaneous Formulations • Pediatric Dosage Handbook • Allen's Compounded Formulations	• www.compoundingtoday.com/ (International Journal of Pharmacy Compounding)
Comprehensive Medical Information	
• The Medical Letter • Treatment Guidelines (from The Medical Letter) • Pharmacist's Letter • Prescriber's Letter	• Pharmacist's Letter (http://www.pharmacistletter.com/) • Prescriber's Letter (https://prescribersletter.com/) • eMedicine (http://emedicine.medscape.com/) • DISEASEDEX (see Table 9.5) • UpToDate (http://www.uptodate.com/) • The Medical Letter (http://www.medicalletter.com/) • Treatment Guidelines (from The Medical Letter) (http://www.medicalletter.com/) • DynaMed (www.dynamed.com/home/) • Disease Management Project: Cleveland Clinic (http://www.clevelandclinicmeded.com/medicalpubs/diseasemanagement/)
Critical Care/Emergency Medicine	
• Hall, Schmidt, and Wood's Principles of Critical Care • Fuhrman and Zimmerman's Pediatric Critical Care • Anesthesiology & Critical Care Drug Handbook (Lexicomp) • Tintinalli's Emergency Medicine	• Principles of Critical Care (via Access Medicine) • Tintinalli's Emergency Medicine: A Comprehensive Study Guide (via Access Medicine) • The Atlas of Emergency Medicine (via Access Medicine)
Dermatology	
• Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology • Lookingbill and Marks' Principles of Dermatology • Clinical Dermatology: A Color Guide to Diagnosis and Therapy	• Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology (via AccessMedicine) • Fitzpatrick's Dermatology in General Medicine (via AccessMedicine)

Continued

TABLE 9.2 List of Tertiary References by Category—cont'd

Books/Printed References	Online References
Drug Identification	
The Pill Book	• Facts and Comparisons eAnswers (see Table 9.3) • Lexi-Drug ID (see Table 9.4) • IDENTIDEX (see Table 9.5) • Pillbox (NLM) (http://pillbox.nlm.nih.gov/index.html) • Drugs.com (http://www.drugs.com/pill_identification.html) • RxList (http://www.rxlist.com/pill-identification- tool/article.htm) • Medscape (http://www.medscape.com/)
Drug Interactions	
• Drug Interaction Facts • Drug Interactions Analysis and Management (Hansten and Horn's) • Stockley's Drug Interactions	• Facts and Comparisons eAnswers (see Table 9.3) • Lexicomp Online (see Table 9.4) • MICROMEDEX (see Table 9.5) • Medscape (http://www.medscape.com/)
Drug Safety	
	• Facts and Comparisons REMS (see Table 9.3) • LexiComp online (see Table 9.4) • BlackBoxRX (www.blackboxrx.com/)
Enteral/Parenteral Nutrition	
• Handbook of Drug Administration via Enteral Feeding Tubes (ASHP) • Clinical Guide to Parenteral Micronutrition	• American Society for Parenteral and Enteral Nutrition (http://www.nutritioncare.org/)
Evidence-Based Medicine	
• Users' Guide to the Medical Literature: Essentials of Evidence-Based Clinical Practice (JAMA Evidence)	• AHRQ (http://www.ahrq.gov/) • Cochrane Library (http://www.cochranelibrary.com/) • National Comprehensive Cancer Network Guidelines (http://www.nccn.org/professionals/physician_gls/f_guidelines.asp) • National Institute for Health and Care Excellence (https://www.nice.org.uk/guidance)
General Pharmacy References	
• AHFS Drug Information (ASHP) • Drug Facts and Comparisons • Drug Information Handbook (Lexicomp) • Pharmacist's Letter	• Facts and Comparisons eAnswers (see Table 9.3) • Lexicomp Online (with AHFS - see Table 9.4) • MICROMEDEX (see Table 9.5) • Pharmacist's Letter (http://www.pharmacistletter.com/) • Hospital Pharmacist's Letter (https://hospitalpharmacistsletter.com/) • Medscape (http://www.medscape.com/) • UpToDate (http://www.uptodate.com/) • NLM Drug Information Portal (http://druginfo.nlm.nih.gov/drugportal/drugportal.jsp)
Hematology/Oncology	
• Wintrobe's Hematology • Drug Information Handbook for Oncology (Lexicomp) • Hematology: Basic Principles and Practice • Cancer: Principles of Practice of Oncology (DeVita's)	• Chemotherapy Resources (see Table 9.3) • PDQ (Physician Data Query—National Cancer Institute) (http://www.cancer.gov/cancertopics/pdq) • MD Anderson Manual of Medical Oncology (via AccessMedicine) • Memorial Sloan Kettering Cancer Center (http://www.mskcc.org/) • Williams Hematology (via AccessMedicine) • National Comprehensive Cancer Network Guidelines (http://www.nccn.org/professionals/physician_gls/f_guidelines.asp)

Continued

TABLE 9.2 List of Tertiary References by Category—cont'd

Books/Printed References	Online References
Herbal Medications and Dietary Supplements	
• Natural Medicines Comprehensive Database • Handbook of Medicinal Herbs (Duke's) • The Review of Natural Products (Facts and Comparisons)	• Natural Medicines Comprehensive Database (http://naturaldatabase.therapeuticresearch.com/home) • Natural Products Database (see Table 9.3) • Office of Dietary Supplements (http://ods.od.nih.gov/) • Dietary Supplements Labels Database (http://www.dsld.nlm.nih.gov/dsld/) • The National Center for Complementary and Integrated Health (https://nccih.nih.gov/) • Memorial Sloan Kettering Cancer Center: About Herbs, Botanicals, and Other Products (http://www.mskcc.org/mskcc/html/11570.cfm)
Immunology and Vaccines	
• Red Book: 2015 Report of the Committee on Infectious Diseases • Pharmaceutical Biotechnology	• Immunization Resources (see Table 9.3) • Review of Medical Microbiology and Immunology (via AccessMedicine) • CDC (http://www.cdc.gov/vaccines) • Red Book: 2015 Report of the Committee on Infectious Disease (http://aapredbook.aappublications.org/)
Infectious Diseases	
• The Sanford Guide to Antimicrobial Therapy • Red Book: 2015 Report of the Committee on Infectious Diseases • Infectious Diseases (Gorbach's) • Principles and Practice of Infectious Diseases (Mandell's)	• Review of Medical Microbiology and Immunology (via AccessMedicine) • Emerging Infectious Diseases (http://www.cdc.gov/ncidod/eid) • Morbidity and Mortality Weekly Report (MMWR: http://www.cdc.gov/mmwr/) • Infectious Diseases Society of America (http://www.idsociety.org/Index.aspx) • Infectious Diseases (see Table 9.4) • AIDSinfo (http://aidsinfo.nih.gov/) • Red Book: 2015 Report of the Committee on Infectious Disease (http://aapredbook.aappublications.org/) • Cleveland Clinic: Guidelines for Antimicrobial Usage 2012–13 (http://www.clevelandclinicmeded.com/medicalpubs/antimicrobial-guidelines/) • John Hopkins Antibiotics and HIV Guides (http://www.unboundmedicine.com/ucentral)
Injectable Drugs	
• Handbook on Injectable Drugs (Trissel's) • King Guide to Parenteral Admixtures • Pediatric Injectable Drugs (Teddy Bear Book) • Gahart's 2017 Intravenous Medications: A Handbook for Nurses and Health Professionals • Extended Stability for Parenteral Drugs (ASHP)	• King Guide (www.kingguide.com) • Trissel's2 (see Tables 9.3–9.5) • Latexdrugs.com/
Internal Medicine	
• Harrison's Principles of Internal Medicine • Conn's Current Therapy	• Harrison's Online (via AccessMedicine)
International Drug References	
• Martindale: The Complete Drug Reference • Index Nominum	• Martindale: The Complete Drug Reference (see Tables 9.3 and 9.4) • ChemIDplus (via TOXNET) • Lexi-Drugs International (see Table 9.4)

Continued

TABLE 9.2 List of Tertiary References by Category—cont'd

Books/Printed References	Online References
Laboratory Information	
• Guide to Diagnostic Tests • Basic Skills in Interpreting Laboratory Data	• Laboratory Values/Body Measurements (see Table 9.3) • Lab Tests and Diagnostic Procedures (see Table 9.4) • Lab Advisor (see Table 9.5)
Law and Regulations	
	• Code of Federal Regulations Title 21 (http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/cfrsearch.cfm)
Nephrology	
• Drug Prescribing in Renal Failure (Bennett's Tables)	• CURRENT Diagnosis & Treatment: Nephrology & Hypertension (via Access Medicine)
Nonprescription Drugs	
• Handbook of Nonprescription Drugs (APhA)	• Facts and Comparisons eAnswers (see Table 9.3) • Lexicomp Online (limited - see Table 9.4) • MICROMEDEX (limited - see Table 9.5)
Nursing and Allied Health	
• 2017 Nursing Drug Handbook • Mosby's (Elsevier) Nursing Skills • Perry and Potter's Fundamentals of Nursing • Perry and Potter's Clinical Nursing Skills &Techniques	• Perry and Potter Clinical Nursing Skills and Techniques (via www.elsevier.com/) • Mosby's (Elsevier) Nursing Skills (via www.elsevier.com/)
Off-Label Information	
• AHFS Drug Information (ASHP) • Off-Label Drug Facts (Facts and Comparisons) • Drug Facts and Comparisons	• Facts and Comparisons eAnswers (see Table 9.3) • Lexicomp Online (limited - see Table 9.4) • MICROMEDEX (limited - see Table 9.5)
Ophthalmic Medications	
• Ophthalmic Drug Facts • The Wills Eye Manual: Office and Emergency Room Diagnosis and Treatment of Eye Disease	
Pain Management	
• Demystifying Opioid Conversion Calculations (ASHP) • Palliative Pharmacy Care (ASHP)	• Hopkins Opioid Program (http://www.hopweb.org/ - free registration) • Facts and Comparisons E Answers—Opioid Agonist Conversion (clinical calculator)
Patient Education	
• Medication Teaching Manual • Pediatric Teaching Manual	• Facts and Comparisons eAnswers (see Table 9.3) • Lexicomp Online (see Table 9.4) • CareNotes (see Table 9.5) • MedlinePlus (http://www.medlineplus.gov/) • KidsHealth (http://kidshealth.org/) • ToxTown (via TOXNET; http://toxtown.nlm.nih.gov/) • PDQ: Physician Data Query (http://www.cancer.gov/cancertopics/pdq)
Pediatrics	
• Pediatric Dosage Handbook (Lexicomp) • The Harriet Lane Handbook • Red Book: 2015 Report of the Committee on Infectious Diseases • Pediatric Pharmacotherapy (ACCP) • Clinical Handbook of Psychotropic Drugs for Children and Adolescents	• Pediatric and Neonatal Lexi-Drugs Online (see Table 9.4) • Current Diagnosis and Treatment: Pediatrics (via AccessMedicine) • KidsHealth (http://kidshealth.org/) • Pediatric Care Online (AAP Textbook of Pediatric Care) (http://www.pediatriccareonline.org/pco/ub/home) • Red Book: 2015 Report of the Committee on Infectious Disease (http://aapredbook.aappublications.org/)

Continued

TABLE 9.2 List of Tertiary References by Category—cont'd

Books/Printed References	Online References
Pharmacogenomics	
• Concepts in Pharmacogenomics (ASHP)	• Pharmacogenomics (see Table 9.4) • The Pharmacogenomics Knowledgebase (https://www.PharmGKB.org/)
Pharmacokinetics	
• Basic Clinical Pharmacokinetics • Applied Biopharmaceutics and Pharmacokinetics (via AccessPharmacy)	• CredibleMeds.org (list of drugs that affects QT interval) • Applied Biopharmaceutics and Pharmacokinetics (via AccessPharmacy)
Pharmacology	
• Goodman and Gilman's: The Pharmacological Basis of Therapeutics	• Goodman and Gilman's: The Pharmacological Basis of Therapeutics (via AccessMedicine)
Pregnancy and Lactation	
• Drugs in Pregnancy and Lactation (Briggs) • Diseases, Complications, and Drug Therapy in Obstetrics: A Guide for Clinicians • Medications and Mother's Milk • Catalog of Teratogenic Agents (Shepard's)	• Pregnancy and Lactation, In Depth (see Table 9.4) • Briggs Drugs in Pregnancy and Lactation (see Tables 9.3 and 9.4) • REPRORISK (see Table 9.5) • LactMed (via TOXNET) • Medications and Mother's Milk (www.medsmilk.com)
Product Availability/Shortages	
• American Drug Index	• CDER (http://www.fda.gov/Drugs/default.htm) ◦ Approved Drug Products with Therapeutic Equivalence Evaluations (Orange Book) ◦ Drugs@FDA Database • FDA Drug Shortages http://www.fda.gov/Drugs/DrugSafety/DrugShortages/default.htm • ASHP Drug Shortages (http://www.ashp.org/shortages)
Product Labeling	
• Physicians' Desk Reference (PDR)	• Drugs@FDA • DailyMed (http://dailymed.nlm.nih.gov/dailymed/index.cfm)
Psychiatry	
• Handbook of Psychiatric Drug Therapy • Diagnostic and Statistical Manual of Mental Disorders (DSM-5) • Clinical Handbook of Psychotropic Drugs for Children and Adolescents	• CURRENT Diagnosis & Treatment: Psychiatry (via Access Medicine)
Therapeutics	
• Goodman and Gilman's: The Pharmacological Basis of Therapeutics • Applied Therapeutics: The Clinical Use of Drugs (Koda-Kimble's) • Pharmacotherapy: A Pathophysiologic Approach (DiPiro's)	• Pharmacotherapy: A Pathophysiologic Approach (DiPiro's) (via AccessPharmacy) • UpToDate (http://www.uptodate.com/) • eMedicine (http://emedicine.medscape.com/) • Merck Manual (http://www.merck.com/mmpe/index.html)
Toxicology, Poisoning, and Environmental Health	
• Goldfrank's Toxicologic Emergencies • Poisoning & Drug Overdose (California Poison Control System) • Tintinalli's Emergency Medicine	• Lexi-Tox (see Table 9.4) • POISONDEX (see Table 9.5) • TOXNET (http://toxnet.nlm.nih.gov/) • Poisoning & Drug Overdose (via AccessMedicine) • Environmental Health and Toxicology SIS (via National Library of Medicine) • OSHA Guidelines • NIOSH Guidelines

Continued

TABLE 9.2 List of Tertiary References by Category—cont'd

Books/Printed References	Online References
Veterinary Information	
• Plumb's Veterinary Drug Handbook • Veterinary Pharmacology and Therapeutics • The Merck Veterinary Manual	• Animal Drugs@FDA (http://www.fda.gov/AnimalVeterinary/default.htm) • DailyMed (http://dailymed.nlm.nih.gov/dailymed/index.cfm)

AccessMedicine (http://accessmedicine.mhmedical.com/), AccessPharmacy (http://accesspharmacy.mhmedical.com/).

topics as nursing (general), patient care, consumer health, alternative medicine, and other health disciplines. Structured subject headings in CINAHL follow MeSH terms and keyword searches, and there are options to limit searches based on the type of journal, publication type, date/range of publication, language, age group, and geographic area of study. This would be a good secondary database to use if one were searching information on patient care issues such as wound care, how to properly move a patient, or how to give an intramuscular injection.

Cochrane Library

The Cochrane Library (www.cochranelibrary.com) has six databases that primarily house systematic reviews and protocols developed by groups and committees within the Cochrane organization. This database is most noted for its systematic reviews which collect and analyze several different trials regarding one particular subject matter and make recommendations for treatment based on the results of the overall analysis. The website can be searched by topic or review group. One might consider this database for searches regarding guidelines, protocols, or thorough reviews of the literature regarding a particular topic (e.g., medications for chemotherapy-induced nausea and vomiting).

EMBASE

EMBASE (www.embase.com) is one of the largest secondary databases available. It allows for access to all the content in MEDLINE, as well as access to more international journals not found in other secondary databases. This would be a go-to database to search for information related to diseases/conditions found in other countries or use of alternative therapies (e.g., traditional Chinese medicine).

Google Scholar

Google Scholar (www.googlescholar.com) is a free, online secondary database that primarily uses keywords (i.e., not MeSH terms) to search for articles. Google Scholar has improved in recent years to eliminate extraneous information not related to published clinical trials; however, there are few limits available to narrow down a search and it may or may not provide access to full-text articles. There may be password restrictions, if one's institution does not subscribe to that particular journal.

International Pharmaceutical Abstracts

Three advantages to IPA are that it is international in its scope, its focus is more about pharmacy and the science of pharmacy, and it contains abstracts/presentations for meetings. IPA is very helpful for searches about drug compounding, intravenous administration rates, and extended stability studies. It is available via EBSCO Health (https://health.ebsco.com/products/international-pharmaceutical-abstracts), but requires a subscription.

PubMed

The National Center for Biotechnology Information, which is part of the US National Library of Medicine (NLM) at the US National Institutes of Health, houses and maintains one of the most commonly used secondary databases in PubMed (https://www.ncbi.nlm.nih.gov/pubmed). It is made up of over 26 million citations from MEDLINE. Citations include those from biomedicine, life sciences, behavioral sciences, chemical sciences, and bioengineering. Free access to the site is available without a need for registration.

TABLE 9.3 Functions Available in Facts and Comparisons eAnswers

Resources Available in Facts and Comparisons eAnswers	
A to Z Drug Facts	Drug Identifier
Boxed Warnings (i.e., Black Box Warnings)	Facts and Comparisons REMS
Briggs Drugs in Pregnancy and Lactation	Immunization Resources IV Compatibility — Trissel's IV Chek
Clinical Calculators	Laboratory Values/Body Measurements
Comparative Data Tables	Manufacturer Index
Don't Crush/Chew List	Natural Products Database (e.g., Review of Natural Products)
Drug and Industry News	Off-Label Drug Facts
Drug Comparison Tables	Patient Assistance Programs
Drug and Disease Interactions Checker	Product Availability
Drug Facts and Comparisons	

TABLE 9.4 Functions Available in Lexicomp Online

Resources Available in Lexicomp Online	
5-Minute Clinical Consult	Lexi-Tox (Poisoning/Toxicology; Nuclear, Biological, and Chemical Agent Exposures)
AHFS DI (Adult and Pediatric)	Martindale: The Complete Drug Reference
AHFS Essentials (Adult and Pediatric)	Natural Products Database
Briggs Drugs in Pregnancy and Lactation	New Drug Reviews
Comparative Efficacy	Patient Education—Adult/Pediatric Medication
Drug Allergy and Idiosyncratic Reactions	Patient Education—Disease and Procedure
Drug Use Evaluations	Pediatric and Neonatal Lexi-Drugs
Geriatric Lexi-Drugs	Pharmacogenomics
Infectious Diseases	Pharmacy and Therapeutics Formulary Reviews
Lab Tests and Diagnostic Procedures	Pharmacy and Therapeutics Summary Reviews
Lexi-Drugs International	Pregnancy and Lactation, In Depth
Lexi-Drugs Online	
Other Useful Tools	
• Charts/Special Documents (under More Clinical Tools—Indexes—specific for each resource)	• Patient Education
• Interaction Checker	• Drug Identification
• IV Compatibility Checker (Trissel's IV Chek)	• Calculators

TABLE 9.5 Functions Available in Micromedex

Resources Available in MICROMEDEX	
Calculators (and other clinically relevant lists)	Lab Advisor (Lab Information)
CareNotes (Patient Education)	Martindale: The Complete Drug Reference
DISEASEDEX Emergency Medicine (Disease Information)	NeoFax
DISEASEDEX General Medicine (Disease Information)	PDR
DRUGDEX Evaluations (Drug Information)	POISINDEX Management (Poisoning and Toxicology)
DRUGDEX Consults (Drug Information)	REPRORISK (Pregnancy and Lactation)
DRUGREAX (Interactions)	• REPROTOX
IDENTIDEX (Drug Identification)	• REPROTEXT
IV Index (Trissels 2 Clinical Pharmaceutics Database)	• Shepard's Catalog
	• TERIS

Primary References

Primary references are simply defined as original research.[32,33,36] Original research is often recognized in the form of a clinical trial, but other forms of original "research" could be case reports, case series, letters to the editor, or conference papers. It is more about being more original than being categorized as bench, lab, or patient research; therefore, something classified as primary literature would depend on the content. For example, review articles, which typically summarize the current literature but do not make original recommendations, are not considered primary literature. They are considered to be tertiary references. Editorials, although they may appear in peer-reviewed journals, may sometimes not be considered primary literature, unless the ideas and observations are completely original and belong only to the author who wrote them. Again, it is up to the reader to interpret the information as to whether or not it is original information; then, it can be classified as primary literature or something else (e.g., tertiary).

The primary advantages of researching the primary literature are that it provides the most up-to-date information on a subject versus tertiary literature. Yet, using primary literature as the sole source for a response may limit the information to one or two particular trials. Other general or background information may need to be found in tertiary literature or review articles.

Most primary references appear in medical journals, which are peer-reviewed. The peer-review process is a review of the proposed publication by experts in that particular field of study.[37] The entire process from first draft to submission may take anywhere between 3 and 12 months but differs between various journals and organizations. However, there are open access journals which may or may not follow the peer-review process. Open access journals may require a fee for publication, but they allow authors who may have been rejected by other journals to publish their results and findings. Therefore, when reviewing open access articles/journals, critical literature evaluation skills are essential to analyze and interpret the data as being useful or not.

Some of the more common peer-reviewed journals are listed here. Many of these journals have other features such as webcasts, podcasts, and videos. Mobile access and tablet applications to journals may also be available. Please check the website for more information or inquire as to mobile/application access via an institutional subscription. In addition, the latest news and articles from many journals can be sent directly to an email account.

New England Journal of Medicine

Renamed the *New England Journal of Medicine* in 1928, the journal can boast that it is the oldest continuously published medical journal, as it has been in circulation since 1812 when it was originally known as the *New England Journal of Medicine and Surgery and the Collateral Branches of Medical Science*. The weekly, peer-reviewed journal is widely international in scope, and access to articles older than 6 months is available for free. The journal and its free-text articles can be accessed at www.nejm.org.

The Lancet

Started in 1823 by an English surgeon, *The Lancet* has expanded into other specialty journals such as *The Lancet Global Health, the Lancet Neurology,* and *The Lancet Infectious Diseases*. The weekly, peer-reviewed journal is one of the best known and most widely used medical journals in the world. The journal can be found at www.thelancet.com, and HINARI-eligible countries (via WHO) may access *The Lancet* free of charge or at a lower cost.

The Journal of the American Medical Association

The Journal of the American Medical Association (JAMA) is a weekly, peer-reviewed, journal published by the American Medical Association. It was started in 1883 and like The Lancet has expanded its readership by including JAMA-specific journals (e.g., JAMA Oncology, JAMA Pediatrics, JAMA Surgery). Access is through www.jamanetwork.com.

Nature

Nature, the parent journal, is a weekly, peer-reviewed publication that was started in 1869 as a "general scientific journal." Today, there are several *Nature* journals specific to topics about chemistry, Earth and the environment, physical sciences, life sciences, and clinical practice. Many of the specific journals (e.g., *Nature Immunology, Nature Chemistry*) are published monthly. Content can be accessed at www.nature.com.

BMJ (Formerly the British Medical Journal)

Started in 1840, *BMJ* is now published online in three editions: a weekly edition for hospital clinicians primarily in the United Kingdom, another weekly edition for general practitioners, and a third monthly edition with a more international audience focused on academic issues. *BMJ* is a peer-reviewed publication and was the first journal to be published completely online in 1995. Access to *BMJ* can be found at www.bmj.com.

Pharmacotherapy: The Journal of Human Pharmacology and Drug Therapy

This monthly, peer-reviewed publication is the official journal of the American College of Clinical Pharmacy (ACCP). Its primary focus is original research articles regarding how medications work in the human body, as well as in-depth review articles on drug therapy. New and unusual case reports of ADRs often appear in this journal. Access can be found at the Wiley Online Library (www.onlinelibrary.wiley.com) by searching *Pharmacotherapy*.

American Journal of Health-System Pharmacy

American Journal of Health-System Pharmacy (AJHP) is the weekly, peer-reviewed publication of the American Society of Health-System Pharmacists (ASHP). One can find drug-related articles about clinical reviews, case reports, and practice research (e.g., MUEs), as well as general pharmacy news about laws, new approvals, and updated guidelines. There is also a Residents' Edition of AJHP that features projects and quality improvement programs completed during pharmacy residency training. The journal can be found at www.ajhp.org.

Drug Information Website Descriptions

Because of the dynamic nature of medical information on the Internet, it would be impractical to include a truly comprehensive listing of all available resources. Utilizing skills in critical appraisal of information sources will ensure that any resource encountered can be properly vetted. However, a short list of reputable and well-organized websites is presented here.

US Food and Drug Administration—*https://www.fda.gov/*

The website of the FDA contains valuable drug information as it pertains to use of medications within the United States. The website contains multiple databases with information specific to product approval and labeling (Drugs@FDA), mandated safety programs (Risk Evaluation and Mitigation Strategies, MedWatch), therapeutic equivalence (Orange Book), and many other resources regarding the clinical and legal use of medications in the United States.

European Medicines Agency—*http://www.ema.europa.eu/ema/*

Similar to FDA, the European Medicines Agency website contains regulatory information regarding the approval and use of medications within the European Union. Information regarding the clinical use and safety of medications as well as herbal products can also be found on this site.

Health Canada Drug and Health Products—*https://www.canada.ca/en/health-canada.html*

Health Canada provides regulatory information regarding the use of medications for Canadian citizens. The site dedicated to drug and health products contains detailed information regarding licensing of health products in Canada, inspections of health products, and alerts based on continual safety monitoring.

Centers for Disease Control and Prevention—*https://www.cdc.gov/*

The website of the Centers for Disease Control and Prevention (CDC) may not initially be thought of as a key place to find drug information. However, many resources contained here provide valuable information regarding the use of medications. The Yellow Book provides information regarding proper vaccination and prophylactic measures that should be considered before traveling to another country. Additionally, vaccination information sheets, vaccine schedules, and guidelines regarding the proper use of vaccines can be found here.

TABLE 9.6 Select Databases Referenced Within the National Library of Medicine Drug Information Portal

MedlinePlus	Consumer-directed health topics; drug information provided by American Society of Health-System Pharmacists
AIDSInfo	Collection of up-to-date guidelines and labeled dosing for medications associated with the treatment of HIV/AIDS
LactMed	Summarized and comprehensive information regarding the use of medications in women who are breastfeeding
LiverTox	Information regarding drug-induced liver injury from a variety of prescription, nonprescription, herbal, or supplement agents
DailyMed	Compilation of FDA-approved drug labels formatted for the Internet
ClinicalTrials.gov	Registry of current and past clinical trials that are both publically and privately supported.
Pillbox	A tool used to aid in the identification of medications
HSDB	Hazardous Substances Data Bank is a toxicology database that provided information regarding data on the toxicity of a chemical including many drugs
PubMed	Secondary database of citations indexed by the US National Library of Medicine
DrugBank	Database of drug information detailed clinical drug data with drug target information on approved products in Canada and the United States
Drugs@FDA	Database of medications approved for use in the United States by the US Food and Drug Administration
European Medicines Agency	Search of the European Medicines Agency website for any drug information regarding human medicines, veterinary medicines, or herbal medicines for human use

Institute for Safe Medication Practices—*https://www.ismp.org/*

Institute for Safe Medication Practices (ISMP) provides a variety of tools to help healthcare institutions increase the safe use of medications. These tools include charts regarding medications that should not be crushed, confused drug names, error-prone abbreviations, high-alert medications, and others. The institute also provides guidelines for healthcare entities to consider in everyday practice such as the safe implementation of smart infusion pumps, practice guidelines for intravenous push medications, and how to create standard order sets. Lastly, they provide several newsletters that bring timely information regarding medication safety to subscribers. ISMP has offices in the United States, Canada, Spain, and Brazil and partners with the International Medications Safety Network.

Drug Information Portal—US National Library of Medicine—*https://druginfo.nlm.nih.gov/drugportal/drugportal.jsp*

The NLM provides a secondary resource through its Drug Information Portal. This resource provides a comprehensive search of drug information that is contained within the wide variety of health information databases of the NLM (Table 9.6).

OBJECTIVE 9.3. OVERVIEW OF DRUG INFORMATION QUALITY ON THE INTERNET

The quantity of medical information and medical literature is growing exponentially. The technology used to access this information is also on the rise. The use of the Internet, smartphones, tablets, and other devices has radically changed the methods by which information is accessed making a wide array of information accessible to many people.[38] However, with this information evolution, the process by which drug information is evaluated has not changed.

The quality of information on the Internet is up for interpretation. In its nature, it is open and, therefore, anyone with the right technical skills could publish a site available to all. Because of this openness, there is the potential for large quantities of information of varying quality that may be unfiltered when an Internet search is performed. This is especially important when considering health information resources on the Internet.[38–41]

Evaluation of Resources

Critical evaluation of drug information websites should entail the same type of process as for other resources such as books or journal articles. One should identify who may gain financially or through reputation from the site, who pays the

TABLE 9.7 Key Considerations in Evaluating Online Health Information

- Authority
  - Are the authors or editors listed?
  - Do those authors specialize in that field?
  - Do they have any concerning conflicts of interest?
  - Is this information associated with a particular institution or group (e.g., American Academy of Pediatrics)?
  - Have there been any published critiques or vetting of the information provided?
- Accuracy
  - Are citations provided?
  - Do the authors properly cite the information provided?
  - Are those references utilized up to date and pertinent to the question at hand?
- Coverage
  - Is the information presented appropriately for the intended audience?
  - Is the information complete consistent and unbiased?
- Currency
  - When was the last revision made?
  - How frequently is the information reviewed and updated?
- Design
  - Is there a logical and efficient organization to the site?
  - Is an index or search function present?
  - Are there pages intended to help navigate the resource?
- Cost
  - Is the cost prohibitive to the type of work that we will be performing?
  - Is enough additional information granted opposed to utilizing free resources?
  - Does a subscription service make the Internet resource better?

expenses for construction and maintenance of the site, what organizations the site may represent, if there is a statement of policy or principles posted on the site, and if there are editors or "reviewers" whose credentials can be checked.

When evaluating an Internet page with drug information, a reader should consider the *authority*, *accuracy*, *coverage*, *currency*, *design*, and *cost* of the resource (Table 9.7).

There are some helpful resources from various organizations to assist in determining how reliable online health information is. A few are listed below:

- US NLM[39]—https://medlineplus.gov/evaluatinghealthinformation.html
- Medical Library Association[41]—http://www.mlanet.org/page/find-good-health-information
- **The DISCERN Instrument** — DISCERN is a brief questionnaire which provides users with a valid and reliable way of assessing the quality of written information on treatment choices for a health problem. DISCERN can also be used by authors and publishers of information on treatment choices as a guide to the standard which users are entitled to expect.[42,43]
- **Health on the Net (HON)** quality label — The Health on the Net Foundation Code of Conduct (HONcode) for medical and health Web sites addresses Internet's main healthcare issues: the reliability and credibility of information. The HON foundation certification focuses only on human health online content.[40]
- **Flesch Reading Ease** — Flesch Reading Ease test rates text on a 100-point scale. It could be found by copying content in a Microsoft Word document. The higher the score, the easier it is to understand the document. For most standard files, the score to be between 60 and 70. The formula for the Flesch Reading Ease score is: $206.835 - (1.015 \times \text{ASL}) - (84.6 \times \text{ASW})$ where: ASL = average sentence length (the number of words divided by the number of sentences) ASW = average number of syllables per word (the number of syllables divided by the number of words).[42,44]

CONCLUSION

Various types of DICs exist, and depending on their set up, each may provide different services to healthcare professionals or the public. However, most DICs have expertise in answering medical- and pharmacy-related questions. DICs have access to a wide variety of tertiary, secondary, and primary databases/sources over and above what individual users can access on their own. These resources are invaluable to the students, residents, and staff of a center when responding to queries related to patients, medications, or therapies, as they allow for the most complete and up-to-date research on the topic at hand. It is important to become familiar with common references/databases to incorporate into the day-to-day practice or research of learners and clinicians.